

Leong Ko Ryan Jasper

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Singapore Citizen — eligible for H-1B1 (US-Singapore FTA), a no-lottery work visa similar to TN

EDUCATION

National University of Singapore

Aug 2023 – Aug 2026

Bachelor of Engineering (Honors), Major in Computer Engineering

Singapore

- GPA: 4.88/5.0 | Dean's List; Enrolled under the Engineering Scholars Program

Stanford University

Apr 2025 – Jun 2025

Visiting Student, Spring Quarter (Exchange)

Stanford, CA

EXPERIENCE

DevOps Engineer

Jan 2025 – Present

ZeroshotData

San Francisco, CA

- Architected and deployed production cloud infrastructure on GCP using Terraform, GKE, Cloud SQL, and VPC, reducing environment setup time from 3 hours to 15 minutes
- Implemented scalable video processing pipelines with GPU-accelerated Kubernetes autoscaling, reducing pipeline costs by 50%
- Configured GitHub Actions CI/CD pipelines and deployed Tailscale mesh VPN with ACL enforcement, replacing Cloudflare tunnels
- Managed infrastructure as code with Terraform, implementing staging/production splits and version-controlled deployments
- Developed and optimized ORB-SLAM3 visual-inertial odometry pipelines, integrating stereo camera feeds and IMU data for real-time robotics navigation
- Engineered EMF tracking system infrastructure, collecting and processing large-scale training datasets for 6-DOF pose estimation models

SWE Intern

May 2024 – Aug 2024

*A*STAR — National Metrology Centre*

Singapore

- Deployed autonomous data collection server on OrangePi 3B+ running Linux, handling driver integration for NI-USB-HS GPIB interfaces and exposing data via a Dash-based web dashboard
- Developed Python middleware to interface with lab instruments, enabling automated data logging and real-time visualization for researchers

SWE Intern

Apr 2023 – Jun 2023

HolyWally

Singapore

- Built internal tools using Python, SQL, and Superblocks with PostgreSQL, integrating with backend APIs and third-party services for ~100 users
- Led QA automation and API integration testing using Postman, collaborating with backend teams to resolve critical UAT bugs for fintech dashboard release

PROJECTS

Flair AI | *AI writing assistant; React, Tiptap, Twilio, Brainbase, Gemini, Tailwind*

2025

- **Winner (1st place, \$1,500)** — AI for Good Hackathon, San Francisco. Built end-to-end application with automated grading and AI voice agent integration for ~400 students
- Architected scalable backend infrastructure using **Twilio**, **Brainbase**, and **Gemini**; built frontend with **React**, **Tiptap**, and **Tailwind CSS**

WatchDocks | *Smart Surveillance System; Flask, YOLOv8, MongoDB, Auth0*

2025

- Developed a real-time surveillance system using **YOLOv8** for object detection and **Flask** for backend API services
- Implemented secure authentication with **Auth0** and data storage with **MongoDB**, ensuring scalable and secure data management

RyNance | *Financial Literacy Platform; Flask, SQLite, Chart.js, Bootstrap*

2022

- Developed full-stack financial literacy platform using **Flask**, **SQLite**, and **Chart.js**; implemented secure user authentication and responsive frontend with **Bootstrap**
- Built interactive tools including retirement calculators and budget planners, deploying the application for public access

SKILLS

DevOps & Cloud: GCP (GKE, Cloud Run, Pub/Sub, Artifact Registry, IAM), Kubernetes, Terraform, Docker, GitHub Actions, CI/CD, Tailscale, Linux

Backend & Systems: Python, Flask, PostREST, Swagger, PostgreSQL, MQTT, C++, Bash Scripting

Tools & Platforms: Git, VS Code, Jira, Confluence, Superblocks, Postman